Lattice refinement followed by the rodeo algorithm: a gate-level benchmark of eigenstate-preparation routes on the Hubbard chain

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Resolution refinement transports an eigenstate adiabatically from a coarse to a fine model space along a shift-controlled, gapped path; followed by the rodeo algorithm, a projective energy filter, it prepares high-fidelity eigenstates without a classical optimizer or training measurements. We implement the lattice version of this pipeline, in which every coarse site is resolved into a block of fine sites, for the Hubbard chain refined from four to eight and from eight to sixteen sites and for the $2 \times 2 \to 4 \times 4$ plaquette, with every stage compiled to rotations and CNOTs and validated against exact diagonalization in the conserved $(N_\uparrow, N_\downarrow)$ sector. Three structural results carry beyond the model. The Galerkin coarse Hamiltonian is itself a Hubbard model with $(t, U) \to (t/2, U/2)$ in one dimension, so the hierarchy is self-similar and can start from the filled dimer, a single basis state. The path conserves total spin and the lattice symmetry, so the transport is governed by the lowest excitation in the symmetry sector of the target, Δ_{eff}, three to four times the sector gap: it sets the refinement time, which does not grow with U/t in the cases studied, and the filter width, and a coarse state with the wrong quantum numbers cannot be refined at all. And the cost of a fixed-step digital filter scales as $1/\Delta_{\text{eff}}$ per cycle with an accuracy floor set by the step. The benchmark is then run on equal terms: the same Trotterized filter is applied to the prolonged and refined states, to an adiabatic ramp from the free-fermion determinant and to single-layer, pruned and adaptively grown (ADAPT-VQE) unitary coupled-cluster states; every route is searched jointly over evolution time, step count and filter cycles, and compared by the minimum number of CNOTs per accepted preparation at a target infidelity. At sixteen and at thirty-two qubits an *unfiltered* ramp from the determinant is the cheapest route from 3×10^{-3} down; the projector path, compiled with its coarse term conditioned on one shared flag qubit, is the cheapest route from the coarse eigenstate, within a factor 2.4–3.1 of the ramp at sixteen qubits and tied with it at 10^{-3} at thirty-two; the adaptive ansatz is cheapest only where it needs no filter cycle. At thirty-two qubits the coarse eigenstate starts ahead of the determinant and transports in a third of the time, but the cheaper step of the ramp cancels the advantage, and a filter cycle buys less than the same gates spent on evolution.

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1 Introduction

Preparing an eigenstate of a many-body Hamiltonian on a quantum computer is the step that every subsequent measurement of a correlated system depends on, and the standard routes each face an obstacle at scale: variational preparation pays for a classical optimization loop and is capped by its ansatz [1, 2], adiabatic preparation pays for the inverse gap along its path [3], and projective filters — phase estimation, polynomial filters, the rodeo algorithm [4, 5] — pay $1/p$ for an input of overlap p with the target. *Resolution refinement* [6] was proposed to supply the filter with a good input without an optimizer: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer model space and transported adiabatically along a path that stays gapped because refinement changes the low-energy spectrum only mildly. In Ref. [7] (paper I below) we implemented this chain — prepare a coarse eigenstate, prolong and refine for a time T, filter with the rodeo algorithm — at the gate level for a hierarchy of single-particle bases of the pairing Hamiltonian, and benchmarked it on equal terms against the same filter applied to cheaper inputs. The verdict there was sober: at eight and sixteen qubits an ordinary adiabatic ramp from the Hartree–Fock determinant reached every target accuracy at a fraction of the cost, and an adaptively grown variational ansatz undercut all filtered routes. The case for the refinement stage rested on its freedom from optimization and on coarse eigenstates that are better starts than the determinant — a regime the pairing hierarchy did not exercise.

The present paper carries the same pipeline, the same benchmarks and the same cost accounting to the Hubbard model [8, 9],

$$H = -\sum_{\langle ij\rangle\sigma} t_{ij}\,(c^\dagger_{i\sigma}c_{j\sigma} + \text{h.c.}) + U\sum_i n_{i\uparrow}n_{i\downarrow}, \quad (1)$$

for three reasons. The Hubbard model is the canonical strongly correlated fermion problem and the most frequently used benchmark for the quantum simulation of fermions [10, 11, 12, 13], with the one-body hopping channel that the pairing model lacks. It is the setting for which resolution refinement was proposed [6]: a *lattice* refinement in which every coarse site is resolved into a block of fine sites, the prolongation is a product of Givens rotations, and the fine lattice is the natural discretization of a continuum problem. And it is not exactly solvable in general, so the exact-diagonalization (ED) reference that anchored every number of paper I must be produced numerically; we keep that discipline and benchmark every stage against ED in the conserved $(N_\uparrow, N_\downarrow)$ sector, up to the 14 400-dimensional sectors of the sixteen-site chain and the 4×4 plaquette.

Relative to the proposal of Ref. [6], the rodeo literature [4, 5, 14, 15] and the related schemes that build large systems from solved pieces — the hierarchical fusion method [16] and nucleation [17], which grows the domain at fixed resolution — this work contributes three things. (a) A *reference implementation* of lattice refinement: the block prolongation, the Galerkin coarse model in closed form, two refinement paths (a shift-controlled projector path and a two-local un-dimerization path), the Trotterized refinement and the Trotterized controlled evolution of the filter, all compiled to rotations and CNOTs, validated against ED at sixteen and at thirty-two qubits and reproducible from one executable notebook [18]. (b) A *benchmark on equal terms*: the refinement-plus-filter chain is compared with the same filter applied to the unrefined prolonged state, to the free-fermion determinant, to an ordinary adiabatic ramp that switches on U from it, and to the single-layer, the pruned and the adaptively grown (ADAPT-VQE [19]) unitary coupled-cluster states, with the ground-state infidelity as the common observable, matched evolution accuracy, a CNOT budget per accepted preparation, and the minimum cost at a given target — at eight sites (sixteen qubits) and, for the first time for this pipeline, at sixteen sites (thirty-two qubits), where the free-fermion input has begun to degrade. (c) *Three results that transfer*: the self-similarity of the Galerkin hierarchy, which lets the chain of refinements start from a single basis state; the identification of the symmetry-protected gap Δ_{eff} as the quantity that controls both the refinement time and the filter width, together with the diagnostics that expose a coarse state that cannot be refined; and the scaling of the fixed-step digital filter — cost per cycle $\propto 1/\Delta_{\text{eff}}$, accuracy floor set by the step — which decides, at every size, whether an input that is cheaper but worse can be rescued by cycles.

The outcome of (b) is stated plainly. At sixteen qubits ($U = 4t$) an unfiltered adiabatic ramp from the free-fermion determinant is the cheapest route to every target accuracy: 1.4×10^4, 2.8×10^4 and 3.7×10^4 CNOTs for infidelities 10^{-3}, 3×10^{-4} and 10^{-4}, against 0.8–2 times that for the adaptive ansatz, 2.4–3.1 times for the projector path with its coarse term conditioned on a shared flag qubit, 3.3–3.6 times for the pruned 98-parameter ansatz and 6–9 times for the two-local refinement path. The reason is specific to the size: the determinant has the same overlap with the target and the same Δ_{eff} as the prolonged coarse eigenstate. At thirty-two qubits the prolonged coarse ground state starts at $p_0 = 0.87$ against 0.72 for the determinant, the projector path transports to 10^{-3} in a third of the ramp's time, the single-layer ansatz has deteriorated by a factor 2.2 and an adaptively grown ansatz of two hundred operators barely reaches 10^{-2}. The projector path then ties with the ramp at 10^{-3} (1.5 against 1.6×10^5 CNOTs) and costs 1.2–1.9 times the ramp at the other targets, the variational inputs are the cheapest at 10^{-2} but need filter cycles below it and then cost four to seven times the ramp, and the two-local path 1.4–6 times. At both sizes the adiabatic routes reach the targets without a filter cycle, whose cost, 2×10^4 and 1.8×10^5 CNOTs, buys more as extra evolution; what decides the comparison is the cost of the transport, and the ramp's step is the cheapest there is.

The paper follows the structure of paper I. Section 2 fixes the model, the sector encoding, the ED reference and the symmetry diagnostics; Sec. 3 defines the lattice prolongation, the Galerkin coarse Hamiltonian, the two paths, the filter and the comparators; Sec. 4 reports the sector-exact results, including the symmetry protection and the wrong-sector plaquette; Sec. 5 gives the circuits and the matched comparison at eight sites; Sec. 6 takes the chain $8 \to 16$ to the circuit level; and Sec. 7 discusses what transfers. The Supplemental Material [20] collects the verification material: the adiabatic bound and the response norm of the path, the dimer-seeded hierarchy and the composition of steps, the plaquette spectra, the variational tables, the Trotterized refinement at eight sites, the exact-ensemble filter, the ingredients of the thirty-two-qubit study and the operating-point tables.

2 Model, reference and symmetry diagnostics

We use $t_{ij} = t = 1$ on nearest-neighbor bonds, open boundary conditions throughout, the chain of L sites and the $L \times L$ square lattice with site index $i = xL + y$. The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$ separately and all calculations are done in one sector of dimension $\binom{N}{N_\uparrow}\binom{N}{N_\downarrow}$.

Encoding and ED. Spin-orbitals are Jordan–Wigner ordered in two blocks, the N up orbitals followed by the N down orbitals. A down hop then picks up the parity of all up electrons twice, which cancels, so the sector Hamiltonian factorizes exactly as

$$H = K_\uparrow \otimes \mathbb{1} + \mathbb{1} \otimes K_\downarrow + U \,\mathrm{diag}(\text{double occupancy}), \tag{2}$$

with K_σ the one-species hopping matrix on the sorted occupation masks of that species. Reference energies and eigenvectors come from dense diagonalization up to a few thousand states and from ARPACK Lanczos beyond; the Kronecker structure is also what makes every product-formula factor of the thirty-two-qubit study a product of two 120×120 exponentials (Sec. 6). The builder is validated against free fermions at $U = 0$, the analytic dimer $E_0 = \frac{1}{2}(U - \sqrt{U^2 + 16t^2})$, and the term decomposition used for product formulas.

Symmetry diagnostics. Three operators are built in the same sector basis and label every eigenstate. The total spin follows from $\langle \mathbf{S}^2 \rangle = S_z(S_z + 1) + \|S^+\psi\|^2$, with $S^+ = \sum_i c^\dagger_{i\uparrow} c_{i\downarrow}$ applied as a map to the $(N_\uparrow + 1, N_\downarrow - 1)$ sector with the Jordan–Wigner signs of the block ordering. A lattice symmetry is a fermionic site permutation, a signed permutation matrix on each species: the reflection parity for the chain, and the eight elements of C_{4v} for the square, from which the weight of a state in each irreducible representation follows from the character formula $w_\Gamma = (d_\Gamma/8) \sum_g \chi_\Gamma(g)^* \langle\psi|g|\psi\rangle$. These labels are what decide, in Sec. 4.2, which gap governs the refinement.

3 The pipeline for lattice refinement

3.1 Prolongation

A refinement step maps a coarse lattice of N_c sites onto a fine lattice of $N_f = bN_c$ sites by resolving every coarse site c into a block $B(c)$ of b fine sites, $b = 2$ for the chain (sites $2c, 2c+1$) and $b = 4$ for the square lattice. The prolongation sends the coarse orbital to the normalized symmetric (bonding) combination of its block,

$$c^\dagger_{c\sigma} \longmapsto s^\dagger_{c\sigma} = \frac{1}{\sqrt{b}} \sum_{f \in B(c)} c^\dagger_{f\sigma}, \tag{3}$$

and a coarse Slater determinant to the product of $s^\dagger$ operators; particle numbers are unchanged, so the filling halves (quarters) at each step, as it must for lattices approaching the continuum at fixed particle number [6]. The map P is an isometry from the coarse into the fine sector; its complement is spanned by configurations in which at least one antibonding block mode is occupied. On a register in which the fine qubits of a block are adjacent, P is one Givens rotation per block per spin for the two-site blocks of the chain, two CNOTs each: 16 CNOTs for $4 \to 8$ and 32 for $8 \to 16$ (a uniform four-site plaquette mode needs three rotations per block and spin). In the sector basis P is built explicitly and $P^\dagger P = \mathbb{1}$ is verified to machine precision.

3.2 The Galerkin coarse Hamiltonian

The coarse Hamiltonian whose eigenstate is refined is the fine Hamiltonian seen at coarse resolution, $H_{\rm low} = P^\dagger H_{\rm high} P$. With this choice $\langle P\psi_{\rm low}|H_{\rm high}|P\psi_{\rm low}\rangle = E_{\rm low}$ holds by construction for the exact coarse eigenstate; it says nothing about the overlap p_0 of the prolonged state with the fine ground state, which must be computed, and it is lost when the coarse state is itself approximate, in which case the coarse error propagates as in paper I — the final infidelity is bounded by $(\sqrt{\epsilon_{\rm low}} + a_{\rm ad} + \eta)^2$ with $a_{\rm ad}$ the adiabatic leakage and η the product-formula error — and, as shown in the Supplemental Material [20], the coarse admixture is in practice transported almost unchanged and caps the refined overlap at roughly the coarse fidelity. The Galerkin model has a closed form. Projecting the hopping term with Eq. (3), an intra-block bond contributes a constant $-t$ per particle on the prolonged subspace and an inter-block bond contributes $\frac{1}{2} s^\dagger_c s_{c'}$; projecting the on-site term, $n_{f\sigma} \to n_{s_c\sigma}/b$, gives $(U/b)\, n_{s_c\uparrow} n_{s_c\downarrow}$. Hence

$$\text{chain:} \quad H_{\rm low} = H_{\rm Hub}(L;\, t/2,\, U/2) - tN, \tag{4}$$

$$\text{square:} \quad H_{\rm low} = H_{\rm Hub}(L \times L;\, t/2,\, U/4) - 2tN, \tag{5}$$

with $N = N_\uparrow + N_\downarrow$ and the boundary conditions of the fine lattice. In one dimension U/t is preserved, so a chain of refinements $L \to 2L \to 4L \to \cdots$ is a sequence of Hubbard chains with the same interaction strength and a hopping scale that doubles at each step: the hierarchy is self-similar, and its coarsest member can be the filled dimer, a single computational basis state. In two dimensions the coarse interaction is weaker by a factor of two relative to its hopping. The Galerkin pair $(H_{\rm low}, H_{\rm high})$ is a choice that defines an algebraic lattice hierarchy at fixed U/t; it is not a continuum limit of one fixed physical model (a contact interaction discretized with spacing a has $t \propto a^{-2}$ and $U \propto a^{-1}$), and a different coarse pair would change not only p_0 but the whole path of Eq. (6), its gaps and its dynamics.

3.3 Two refinement paths

Path A: the shift-controlled projector path. As in paper I,

$$H_A(s) = \cos^2\!\left(\tfrac{\pi s}{2}\right) P(H_{\rm low} - \mu)P^\dagger + \sin^2\!\left(\tfrac{\pi s}{2}\right)(H_{\rm high} - \mu), \qquad s = t/T, \tag{6}$$

with the shift $\mu = E_{\rm low} + \mu_{\rm buf}$ placing the complement of the prolonged subspace, where the coarse term vanishes, at energy $\mu_{\rm buf}$ above the tracked state at $s = 0$. Because $P(H_{\rm low} - \mu)P^\dagger = \Pi_a(H_{\rm high} - \mu)\Pi_a$ with Π_a the projector onto empty antibonding modes, the coarse term is, on a device, the fine Hamiltonian in the bonding/antibonding frame conditioned on all antibonding qubits being in $|0\rangle$ — the multi-controlled structure whose cost dominated the $3 \to 4$ chain of paper I, now with $2N_c$ control qubits.

Path B: adiabatic un-dimerization. The same end points are connected by a path that stays two-local in the laboratory frame: add to

the fine Hamiltonian an extra hopping Λ on the intra-block bonds and switch it off,

$$H_B(s) = H_{\text{high}} - \Lambda \cos^2\!\left(\tfrac{\pi s}{2}\right) \times \sum_{\langle f f' \rangle \in \text{intra}} \sum_{\sigma} \left(c^\dagger_{f\sigma} c_{f'\sigma} + \text{h.c.}\right). \quad (7)$$

At $s = 0$ the antibonding modes are pushed up by 2Λ; for $\Lambda \to \infty$ the ground state of $H_B(0)$ tends to the prolonged Galerkin ground state (the initial Hamiltonians and exact initial eigenstates of the two paths coincide only in that limit), and at finite Λ it differs from it by an admixture of order t/Λ, so a transport that starts from the prolonged state carries a residual $1 - p \simeq c\,(t/\Lambda)^2$ to the end. Path B is the dimerized Hubbard chain being made uniform; every Trotter step along it costs what a step of the fine Hamiltonian costs, with no conditioning.

Refinement time. The exact evolution along either path splits the s interval into M_s slices (96 up to 784 states, 48 with a sparse Krylov exponential for the 14 400-dimensional sectors), diagonalizes or applies the Hamiltonian at the slice midpoint, and reports the overlap $p(T) = |\langle \Psi_{\text{high}} | \psi(T) \rangle|^2$ with the ED ground state on a grid of T; T^* is the first grid value with $1 - p < 10^{-3}$. As in paper I the pipeline is optimizer-free in execution but not parameter-free: T, μ_{buf} (or Λ), the number of product-formula steps n_s, the filter step δt and the objective energy E_{obj} are all chosen with knowledge of the exact answer, and a device-level calibration protocol for them is part of the open work.

3.4 The rodeo filter

The filter is unchanged from paper I: M cycles of the controlled evolution $e^{-iH_{\text{high}}t_k}$ on one reused ancilla with the phase $e^{iE_{\text{obj}}t_k}$, Gaussian cycle times of RMS width σ, and post-selection on the ancilla. For an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ and $E_{\text{obj}} = E_0$ the fidelity of the ensemble of accepted runs obeys exactly

$$1 - \mathcal{F}_M = \frac{\sum_{k \neq 0} |c_k|^2 q_k^M}{p + \sum_{k \neq 0} |c_k|^2 q_k^M}, \qquad q_k = \tfrac{1}{2}\left(1 + e^{-(E_k - E_0)^2 \sigma^2 / 2}\right), \quad (8)$$

the average over the Gaussian ensemble of cycle times with exact controlled evolution and E_{obj} on resonance; the factor $q_k^M \geq 2^{-M}$ makes 2^{-M} a lower bound on the suppression of each excited component, and a sampled average over a finite number of time sets, as in the circuit-level studies below, scatters around Eq. (8) with a relative standard error that grows with M. The width is set by the smallest excitation that carries weight in the input, $\sigma = 5/\Delta_{\text{eff}}$ (Sec. 4.2); choosing it from the bare sector gap would lengthen the controlled evolutions by a factor of up to three (sixteen sites) for no gain.

3.5 The comparators

Four inputs that need no coarse model are fed to the same filter. The *free-fermion determinant* (the N_σ lowest orbitals of the hopping matrix) is prepared by a Givens network of $N(N-1)/2$ rotations per spin. The *ordinary adiabatic ramp* switches on the interaction from it, $H_R(s) = K + \sin^2(\pi s/2)\, U \sum_i n_{i\uparrow} n_{i\downarrow}$; its product-formula version is the fine-chain step with a rescaled interaction angle, with no shift and no coarse term — the analog of the Hartree–Fock ramp of paper I and the baseline every refinement path must beat. The *unitary coupled-cluster singles-and-doubles* (UCCSD) state is the disentangled product $\prod_\mu e^{\theta_\mu(\tau_\mu - \tau_\mu^\dagger)} |\text{HF}\rangle$ of the spin-resolved singles and doubles from the determinant, in a fixed order (all singles, then the doubles, each class in lexicographic order; a different ordering is a different ansatz), optimized by L-BFGS with the analytic gradient; the simulation is exact for the ansatz, every generator being a Givens-type rotation $1 + \sin\theta\, K + (\cos\theta - 1)\Pi$ in the orbital occupation basis. Since most amplitudes are small we also report the *pruned* ansatz obtained by keeping the operators whose converged amplitude exceeds 10^{-3} and re-optimizing. And *ADAPT-VQE* grows the ansatz from the same pool one operator at a time, appending at every iteration the generator with the largest energy gradient (repeats allowed) and re-optimizing all amplitudes warm-started from the previous ones, so that every prefix of the sequence is itself a converged state; this is the adaptive construction that the pruned ansatz approximates. The simulated state is $\Phi\, U_{\text{orb}} |\text{ref}\rangle$ with Φ the orbital-to-site rotation, so on a device the excitation blocks act in the orbital basis on the reference occupation and the Givens network Φ is applied once, last; we count them with the Pauli-rotation prim-

itive of Sec. 5: a single compiles into two strings of weight $a - i + 1$, a double with sorted Jordan–Wigner indices $p < q < r < s$ into eight strings of weight $(q - p + 1) + (s - r + 1)$ (the Z strings cancel between q and r), a weight-w string costs $2(w - 1)$ CNOTs.

4 Sector-exact results

4.1 The chain refined from four to eight sites

Table 1 and Fig. 1 summarize the refinement of the four-site chain with $N_\uparrow = N_\downarrow = 2$ (36 states) to the eight-site chain (784 states) for $U/t = 0, 2, 4, 8$. Three features stand out. (i) The prolonged state is a good start at every coupling, $p_0 = 0.75$–0.81, with a weak maximum at $U = 2t$; unlike the pairing model, the free-fermion case is not trivial, because the bonding modes are not the lowest fine orbitals. (ii) The sector gap of the fine Hamiltonian falls by a factor of four over the range of U, from $0.53t$ to $0.14t$, but the refinement time does not move: with $\mu_{\mathrm{buf}} = 2t$ the overlap reaches $p > 0.997$ at $T = 8/t$, $1 - p < 10^{-3}$ at $T^* = 10/t$ and $p > 0.9999$ at $T = 16/t$ for all four couplings, and with $\mu_{\mathrm{buf}} = t$ at roughly twice the time. The minimum of the sector gap along Path A sits at $s = 1$ (it is the fine-sector gap itself) in every case, so the global path gap cannot be what controls the transport. (iii) Path B refines at a comparable rate but saturates at the finite-Λ ceiling: at $T = 64/t$, $1 - p = 4.8$–7.3×10^{-3} for $\Lambda = 8t$ and 1.7–3.7×10^{-3} for $\Lambda = 16t$, consistent with the $(t/\Lambda)^2$ admixture (the prolonged state's overlap with the exact ground state of $H_B(0)$ is 0.994–0.995 at $\Lambda = 8t$ and 0.998–0.999 at $16t$). The ceiling, not the transport, limits Path B.

4.2 Which gap controls the refinement: symmetry protection

Both terms of Eq. (6) commute with the total spin and with the point group of the fine lattice (the block structure respects the reflection of the chain and the C_{4v} group of the square), and the prolongation maps irreducible representations onto themselves. The transported state therefore never leaves the symmetry sector of the coarse ground state — a spin singlet, even under reflection, for Table 1 — and the states that can be excited along the path are only those of the same sector. Figure 2 makes this explicit: for $U = 4t$ the lowest fine excitations at $0.223t$ (a reflection-odd triplet) and $0.459t$ (an even triplet) carry weights of order 10^{-31}, and the first state with nonzero weight (4.6×10^{-5}), an even singlet, lies at $\Delta_{\mathrm{eff}} = 0.588t$; at $U = 8t$ the corresponding numbers are $0.137t$ and $0.286t$ (triplets, zero weight) and $\Delta_{\mathrm{eff}} = 0.366t$. Δ_{eff} falls with U much more slowly than Δ, which explains the U-independent T^*.

Figure 3 follows two gaps along Path A: the sector gap and the transport gap, the gap to the lowest state with the same (S, parity) as the tracked state within the invariant sector of the path. The latter never approaches the former; its minimum is Δ_{eff} itself, at $s = 1$. (The transport gap of the path and the input-support gap of a filter input — the smallest final excitation carrying weight in that particular state — are distinct quantities that coincide for the inputs used here; a spin-resolved UCCSD state is not automatically $\mathbf{S}^2$- or parity-adapted, and we checked that the full and pruned UCCSD states carry only $\sim 10^{-7}$ weight below Δ_{eff}.) This is the isolated-branch condition of the adiabatic theorem verified numerically along the whole path, within the symmetry sector in which the transport actually takes place. The sufficient adiabatic time of Jansen, Ruskai and Seiler [21], evaluated as in paper I once with the sector gap and once with the symmetry-resolved gap, is smaller by a factor 5 to 17 in the second form but remains three to five orders of magnitude above the empirical $T^* = 10/t$ (Supplemental Material [20]); the slack is that of conservative norm estimates and of $\epsilon = 10^{-2}$ against an empirical 10^{-3}, not of different criteria, since for a rank-one branch the projector distance of the bound is the infidelity. The first-order response norm of the path, $\chi_u(s)$ — the squared eigenvector response along the affine path in u, a sensitivity diagnostic rather than the leakage itself, which also depends on the schedule and on the oscillatory phases set by T — is largest at the initial endpoint for every coupling, $\chi_u(0) = 0.38$–0.87 for $U/t = 0$–8, and its modest growth with U is consistent with the U-independent T^* (Supplemental Material [20]).

Table 1: Refinement of the open Hubbard chain from $L = 4$ to $L = 8$ sites at $N_\uparrow = N_\downarrow = 2$ ($t = 1$; 36 and 784 states). E_{low} is the ground-state energy of the Galerkin coarse model, Eq. (4), E_{high} the ED ground-state energy of the fine chain, p_0 the weight of the prolonged coarse ground state on the fine ground state, Δ the sector gap of the fine Hamiltonian and Δ_{eff} the lowest excitation energy carrying weight in the prolonged state. $p(T)$ is the overlap after refinement along Path A ($\mu_{\mathrm{buf}} = 2t$) and Path B ($\Lambda = 16t$); the last two columns give the ensemble infidelity, Eq. (8), of the rodeo filter with $\sigma = 5/\Delta_{\mathrm{eff}}$ after six cycles, applied to the prolonged state and to the Path-A state refined for $T = 16/t$.

U/t	E_{low}	E_{high}	p_0	Δ	Δ_{eff}	$p_A(8)$	$p_A(16)$	$p_B(16)$	$p_B(32)$	$1-\mathcal{F}_6$, prol.	$1-\mathcal{F}_6$, ref.
0	-6.236068	-6.822948	0.777	0.532	0.879	0.99718	0.99997	0.99660	0.99838	4.5×10^{-3}	5.4×10^{-7}
2	-5.437971	-6.032090	0.814	0.319	0.780	0.99876	0.99996	0.99594	0.99849	3.6×10^{-3}	5.9×10^{-7}
4	-4.976573	-5.634427	0.797	0.223	0.588	0.99837	0.99994	0.99316	0.99769	4.0×10^{-3}	9.2×10^{-7}
8	-4.558586	-5.277973	0.750	0.137	0.366	0.99761	0.99991	0.99205	0.99731	5.2×10^{-3}	1.4×10^{-6}

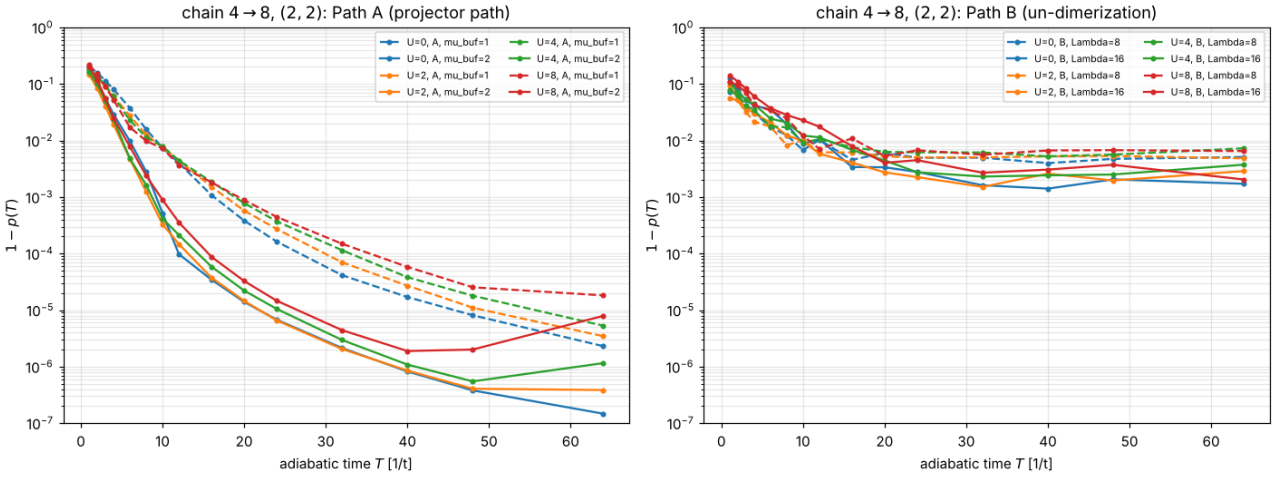


Figure 1: Refinement of the open chain $4 \to 8$ at $(2,2)$: $1-p(T)$ along Path A (left; solid $\mu_{\mathrm{buf}} = 2t$, dashed $\mu_{\mathrm{buf}} = t$) and along Path B (right; solid $\Lambda = 16t$, dashed $\Lambda = 8t$) for $U/t = 0, 2, 4, 8$. Path A reaches $1-p < 10^{-4}$ by $T = 16/t$ at every coupling; Path B saturates at the finite-Λ ceiling of a few 10^{-3}.

4.3 The self-similar hierarchy: from the filled dimer to sixteen sites

The self-similarity of Eq. (4) allows the chain of refinements to start from the two-site dimer with $N_\uparrow = N_\downarrow = 2$, whose sector contains a single state — both sites doubly occupied — with $E_{\mathrm{low}} = U - 4t$: the analog of the $k = 2$ determinant that seeded the $2 \to 4$ chain of paper I, requiring no classical or variational preprocessing. The step $2 \to 4$ starts from $p_0 = 0.80$ at $U = 0$ down to 0.48 at $U = 8t$, but with a large effective gap ($\Delta_{\mathrm{eff}} = 2.2t$–$0.8t$) it reaches $p > 0.9997$ by $T = 16/t$ at every U. Composing steps with the *refined* four-site state as the input of the $4 \to 8$ step, and the shift referenced to its variational energy, the chain $2 \to 4 \to 8$ at $U = 4t$ with $T = 16/t$ per step reaches $p = 0.99988$, against 0.99994 from the exact coarse input; with $T = 8/t$ in the first step the second saturates at $p = 0.9979$, the overlap of its input, because the admixture of other coarse states refines to other fine states. The composed chain also beats a single-shot refinement $2 \to 8$ with blocks of four ($p_0 = 0.579$, $p = 0.9966$ at $T = 16/t$). The protection survives a doubling of the system: refining the eight-site chain to sixteen sites at $(2,2)$ (784 $\to$ 14 400 states, $U = 4t$) starts from $p_0 = 0.869$, faces a fine-sector gap of only $0.051t$, and still reaches $p = 0.9924$ at $T = 8/t$, 0.99931 at $T = 16/t$ and 0.99995 at $T = 32/t$, because the first state to carry weight lies at $\Delta_{\mathrm{eff}} = 0.138t$ while the eight other states below $0.4t$, including the one that defines the sector gap, carry none. Three steps from the filled dimer, $2 \to 4 \to 8 \to 16$ with $T = 16/t$ each and the refined state as the input of every step, give $p = 0.99923$ for the sixteen-site ground state. The tables and figures of the dimer step and of the composed chains are in the Supplemental Material [20].

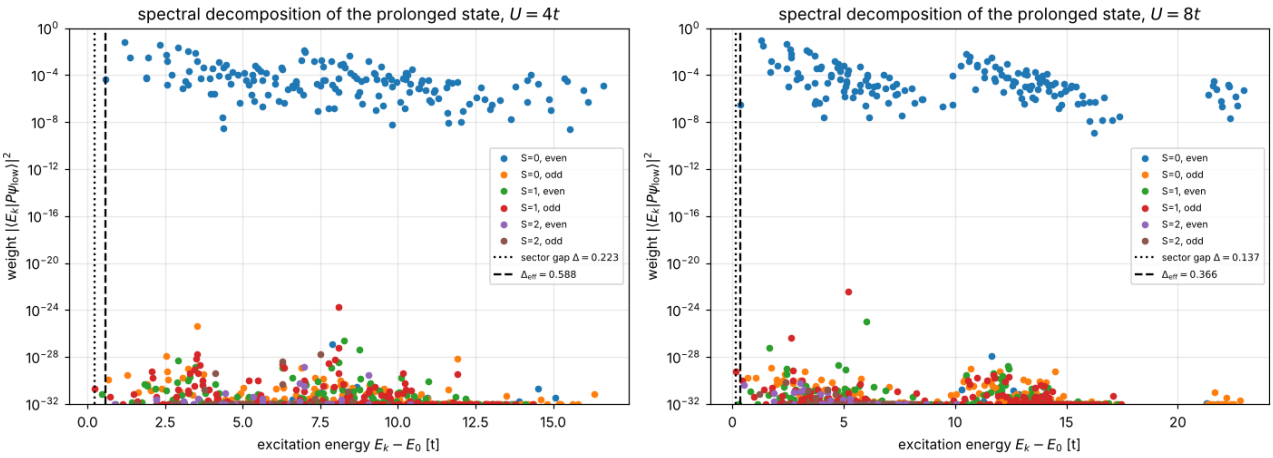


Figure 2: Spectral decomposition of the prolonged coarse ground state over the eigenstates of the eight-site chain at $U = 4t$ (left) and $8t$ (right), each eigenstate labeled by its total spin and reflection parity. Only even singlets carry weight; the triplets and odd states, including the ones that define the sector gap Δ (dotted), sit at round-off level. The lowest state with weight defines $\Delta_{\rm eff}$ (dashed).

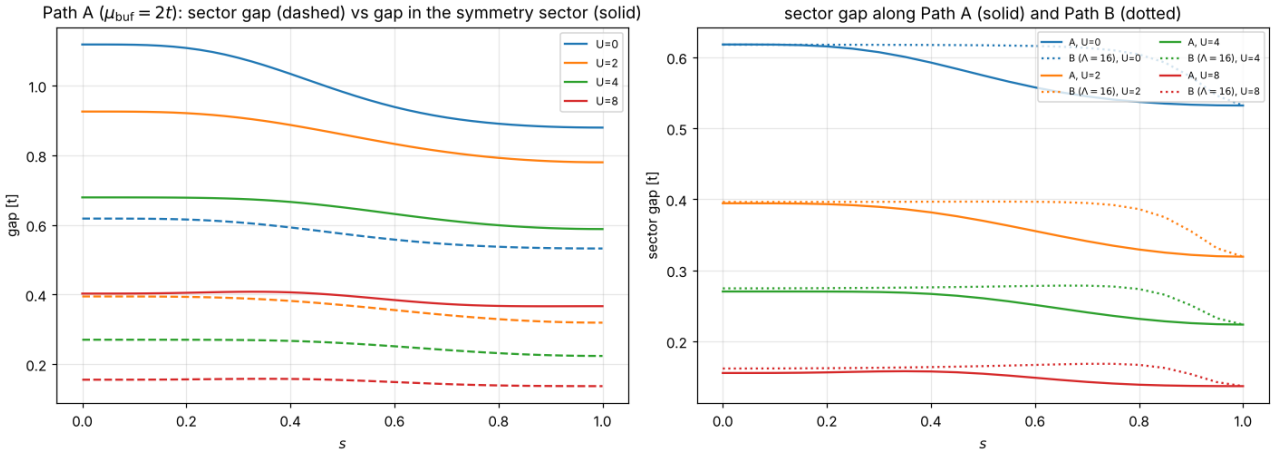


Figure 3: Left: along Path A ($\mu_{\rm buf} = 2t$), the sector gap (dashed) and the gap within the symmetry sector of the tracked state (solid) for $U/t = 0, 2, 4, 8$. Right: the sector gap along Path A (solid) and Path B with $\Lambda = 16t$ (dotted); both reach their minimum, the fine-sector gap, at $s = 1$.

4.4 Two dimensions: the plaquette and the wrong symmetry sector

For the square lattice the first step is $2 \times 2 \to 4 \times 4$. With one particle per spin ($16 \to 256$ states) the refinement behaves as in one dimension: both ground states are A_1, $p_0 = 0.80$–0.85 for $U/t = 0, 4, 8$, $\Delta_{\rm eff} \simeq 2.1t$–$2.2t$, and $p > 0.9997$ at $T = 8/t$. With two particles per spin ($36 \to 14\,400$ states) the prolonged coarse ground state has *zero* overlap with the fine ground state, and the symmetry diagnostics identify why (Fig. 4). The half-filled coarse plaquette (Galerkin parameters $t/2$, $U/4$) has, at every $U/t = 1$–8 studied, a B_1 singlet ground state with an A_2 triplet close above it ($0.071t$ at $U = 4t$); the dilute fine lattice, four electrons on sixteen open sites, has an A_2 *triplet* ground state for every $U > 0$ studied ($U/t = 1$–8), the B_1 singlet lying $0.04t$–$0.11t$ above it. The prolongation commutes with $\mathbf{S}^2$ and with C_{4v}, so the two cannot be connected: the fine ground state has weight 0 on the prolonged coarse ground state, 0.647 on the prolonged A_2 triplet and 0.007 on all other coarse states together ($U = 4t$). Refining that triplet along Path A ($\mu = E_1^{\rm low} + 2t$) reaches $p = 0.99944$ at $T = 8/t$ and 0.99997 at $T = 16/t$ although the fine-sector gap is only $0.090t$, since the lowest state of the $(A_2, S = 1)$ sector that carries weight lies $1.77t$ above the ground state; the same path started from the coarse ground state gives

$p = 3 \times 10^{-27}$. The lesson for the method is that the state to refine is the lowest coarse eigenstate *in the symmetry sector of the target*, which need not be the coarse ground state, that the coarse solver must be run with the target's quantum numbers imposed or at least diagnosed, and that the filter cannot rescue a wrong choice, since a state of zero overlap has nothing to amplify. Conversely the refinement is a sensitive detector of a level crossing between the coarse and the fine problem — here a singlet–triplet crossing as the filling goes from one half to one eighth, a property of the filling, not of U, over the whole range studied. The spectra and the table of the plaquette cases are in the Supplemental Material [20].

4.5 Filtering, and the variational inputs

With $\sigma = 5/\Delta_{\text{eff}}$ ($5.7/t$ to $13.6/t$ for $U/t = 0$ to 8) the exact ensemble filter, Eq. (8), applied to the prolonged state takes the infidelity from $1 - p_0 \simeq 0.2$ to $4\text{–}5 \times 10^{-3}$ after six cycles and $2\text{–}3 \times 10^{-4}$ after ten, with an acceptance that converges to p_0; applied to the Path-A state refined for $T = 16/t$ it starts from 10^{-4} and reaches $0.5\text{–}1.4\times10^{-6}$ after six cycles; applied to the eight-site UCCSD state it starts from 4.7×10^{-4}, 3.1×10^{-3} and 1.2×10^{-2} at $U/t = 2, 4, 8$ and reaches 7×10^{-6}, 5×10^{-5} and 2×10^{-4} (figure in the Supplemental Material [20]). The refined input converges faster per cycle by the ratio of the initial infidelities; whether the cycles saved pay for the refinement gates is the circuit-level question of Sec. 5.

The variational tables are in the Supplemental Material [20]; what matters here is three numbers at $U = 4t$. The 198-parameter single-layer UCCSD on the eight-site chain reaches $1 - F = 3.1 \times 10^{-3}$ at 3.1×10^4 CNOTs; 98 of its amplitudes exceed 10^{-3}, and keeping those (12 singles, 86 doubles) and re-optimizing gives the same accuracy, 3.145×10^{-3} against 3.143×10^{-3}, at 1.5×10^4 CNOTs. Used as the coarse solver on the four-site chain, the 26-parameter ansatz ($1 - F = 3.3 \times 10^{-3}$) prolongs with the same p_0 as the exact coarse state (0.800 against 0.797) but the refinement then saturates at $p = 0.9967$ instead of 0.9999: the coarse stage must be as accurate as the target, or the filter must do the rest.

5 Circuits and the matched comparison at eight sites

5.1 Pauli structure and step costs

In the block Jordan–Wigner ordering an open chain of L_f sites on $2L_f$ qubits has $4(L_f - 1)$ hopping strings $\frac{t}{2}(X_jX_{j+1} + Y_jY_{j+1})$ of weight two, L_f interaction strings $\frac{U}{4}Z_jZ_{j+L_f}$ and $2L_f$ single-Z terms: 52 strings of weight ≤ 2 for $L_f = 8$, verified against the full-space Pauli decomposition on eight qubits, against the 325 strings of weight up to eight of the $k = 4$ pairing model on half the qubits. With the Pauli-rotation primitive of paper I — $2(w - 1)$ CNOTs per weight-w string, two more per rotation when controlled — a second-order Trotter step of the eight-site chain costs 144 CNOTs and a controlled one 288; at sixteen sites, 304 and 608 (at fixed particle number the sum of the $2L_f$ single-Z terms is a scalar, so in the controlled step it is one phase on the ancilla, absorbed into the objective-energy phase gate; charging it would give 352 and 736). The product formula groups the bonds into two layers of disjoint bonds plus the diagonal interaction; each layer is a product of commuting two-qubit blocks and is realized exactly, so the Trotter error comes only from the non-commutativity between the three groups, as on hardware. The prolongation $4 \to 8$ costs 16 CNOTs. The embedded coarse term of Path A is the coarse Hubbard step on the bonding register (24 rotations, single-qubit phases included since conditioned they are no longer free) with every rotation conditioned on the eight antibonding qubits being empty, plus the conditioned scalar phase of the constant part of $H_{\text{low}} - \mu$. The coarse evolution never changes the control qubits, so their conjunction is computed once into a clean flag qubit with a ladder of relative-phase Toffolis ($6(c - 1)$ CNOTs for the ladder and its uncomputation), every rotation is singly controlled by the flag (2 CNOTs) and the flag is uncomputed before the next fine block: 124 CNOTs per application, plus the frame change into and out of the bonding/antibonding basis (2×16 CNOTs, the prolongation network and its inverse, unless merged with a neighboring block), i.e. 156 CNOTs or 1.1 fine steps, at the price of $c - 1 = 7$ clean work qubits beyond the $2L_f + 1$ of the rest of the circuit; at sixteen sites (16 controls, 52 rotations) $268 + 64 = 332$, 1.1 fine steps and 15 work qubits. A ladder per rotation, the con-

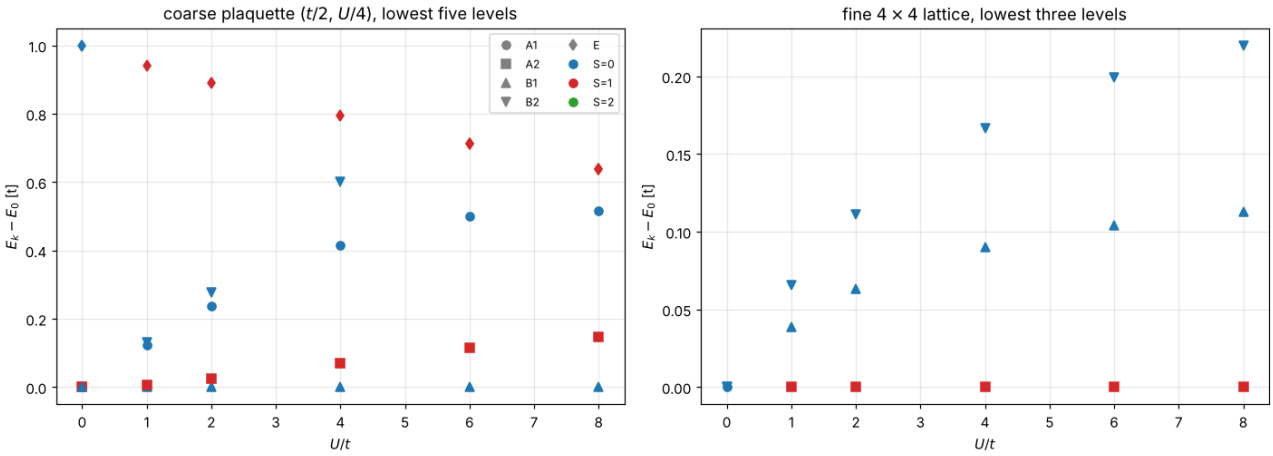


Figure 4: Lowest levels of the coarse plaquette (left, Galerkin parameters $t/2$, $U/4$) and of the fine 4×4 lattice (right) at $(2, 2)$ versus U/t, labeled by C_{4v} irrep (marker) and total spin (color). For every U/t studied the coarse ground state is a B_1 singlet and the fine ground state an A_2 triplet; the prolongation conserves both labels.

vention of an earlier version of this work, would cost 1 132 and 4 948 CNOTs per application, and the Gray-code phase polynomial of paper I is out of the question at this number of controls.

5.2 Trotterized refinement and the baselines

With second-order product formulas over the groups $[P(H_{\text{low}} - \mu)P^\dagger$ split into its three conditioned coarse groups, $K_{\text{even}}, K_{\text{odd}}, D - \mu]$ for Path A and $[(1 + \Lambda \cos^2)K_{\text{even}}, K_{\text{odd}}, D]$ for Path B, the exact-slice overlap at $T = 16/t$, $U = 4t$ is reproduced to within 10^{-3} with $n_s = 128$ steps for Path A ($p = 0.99982$), 256 for Path B at $\Lambda = 8t$ (0.99213) and 512 at $\Lambda = 16t$, against 43 steps for the pairing path of paper I: the shifted generators have larger norms ($\|B - A\| = 14$ for Path A, $1 + \Lambda$ on the intra-block bonds for Path B), which makes $\Lambda = 8t$ the better operating point of the two-local path despite its higher ceiling. At these step counts the refinement costs 5.8×10^4 CNOTs for Path A (4.0×10^4 in the coarse term with the shared-flag compilation; 3.0×10^5 with a ladder per rotation) and 3.7×10^4 for Path B, both from an exact coarse eigenstate. The ramp from the free-fermion determinant (Givens network 112 CNOTs) has $p_0 = 0.7804$ — essentially the overlap of the prolonged state, 0.7974 — and the same $\Delta_{\text{eff}} = 0.588t$, since it lies in the same symmetry sector; with exact slices it reaches $p = 0.99686$, 0.99960 and 0.99989 at $T = 8, 12, 16$, as fast as Path A and far above the ceiling of Path B, and its product-formula version needs 64 steps at $T = 8/t$ (9 328 CNOTs) and 128 at $T = 16/t$ (0.999659, 18 544 CNOTs). The step scans are in the Supplemental Material [20].

5.3 Matched-observable comparison

Following Sec. V.C of paper I we feed the same Trotterized rodeo filter — second order, fixed step δt, $\sigma = 5/\Delta_{\text{eff}} = 8.5/t$, up to six cycles — to every input and compare the ensemble infidelity of the accepted runs against the CNOTs per accepted preparation (input plus filter, early termination after the first failed ancilla measurement). The metric is the amortized logical CNOT count per accepted state on an ideal, all-to-all-connected device — not the total work including the rejected attempts, and not a scheduled depth — for a register of $2L_f + 1$ qubits ($2L_f + c$ for the shared-flag coarse term of Path A). Two implementation details are forced by the Hubbard norms. The fixed step is diagonalized once, so that n steps are applied exactly and the cycle times are quantized to multiples of δt (on average 68 steps per cycle, 2.0×10^4 CNOTs); the filter then projects on the eigenvector of the step unitary closest to the ground state, whose eigenphase is what the energy scan of the rodeo locates on a device, and we take E_{obj} equal to it rather than to E_0. The Trotter floor of the filter is that eigenvector's infidelity with the exact ground state: 2.6×10^{-3} at $\delta t = 0.25/t$ (the step of paper I, where $E_{\text{obj}} - E_0 = 0.09t$ would also have spoiled the acceptance), 5.8×10^{-5} at $\delta t = 0.1/t$ and 3.5×10^{-6} at $0.05/t$; we operate at $\delta t = 0.1/t$, where $E_{\text{obj}} - E_0 = 0.013t$. Because the cycle

times are quantized every cycle shares the eigenvectors of the step unitary, so the long-run state of the filter is that eigenvector itself and, with E_{obj} on its eigenphase, the acceptance tends to the input's weight on it instead of decaying — the fixed-step analog of the averaged post-selection map of paper I. The same structure makes the ensemble of accepted runs *exactly* computable at this size: in the eigenbasis of the step unitary the accepted density matrix after M cycles is $\rho_{ij}^{(M)} = c_i c_j^* Q_{ij}^M$ with $Q_{ij} = \mathbb{E}_n[f_i(n) f_j(n)^*]$, $f_j(n) = \frac{1}{2}[1 + e^{in(\phi_j + E_{\mathrm{obj}}\delta t)}]$ and n the rounded Gaussian step count, so no schedule sampling enters the eight-site numbers (the exact ensemble also corrects a sampled value of an earlier version that had wrongly passed a target: the $T = 8/t$ ramp with 64 steps reaches 1.05×10^{-4} after six cycles, not 8.9×10^{-5}).

The inputs are not a fixed list but a search. The ramp is scanned over $T = 8$–$48/t$ and $n_s = 64$–512, Path B over $\Lambda = 8t$, $16t$, $32t$, $T = 16$–$32/t$ and $n_s = 128$–$1\,024$, Path A (shared-flag coarse term) over $T = 8$–$16/t$ and $n_s = 64$–512, each with $M = 0, \ldots, 6$ cycles, and the adaptive ansatz at every prefix; the coarse state of the hierarchy is not an oracle but the state actually prepared by a four-site ramp from the free-fermion determinant, in two versions — 8 216 CNOTs for $1 - F = 6.7 \times 10^{-5}$ and 16 408 for 1.4×10^{-5} — that are prolonged ($p_0 = 0.7954$ and 0.7967 against 0.7974 for the exact coarse eigenstate) and transported, so that the search can trade coarse accuracy against filter cycles. Table 2 gives the minimum cost per route at the three targets and Fig. 5 the lower envelopes.

The verdict at eight sites is read from Table 2. The cheapest route at every target is an *unfiltered* adiabatic ramp from the free-fermion determinant: 1.4×10^4, 2.8×10^4 and 3.7×10^4 CNOTs for 10^{-3}, 3×10^{-4} and 10^{-4} — a filter cycle costs 2×10^4 CNOTs, as much as doubling the step count of a $T = 16/t$ ramp, and the finer or longer unfiltered evolution buys more. The adaptive ansatz is the closest competitor and the only one that beats the ramp anywhere: 1.7×10^4 CNOTs at 10^{-3} and 2.3×10^4 at 3×10^{-4} (149 and 194 operators, no cycle), 0.8 times the ramp at the second target, but it needs cycles below 3×10^{-4} and costs twice the ramp at 10^{-4}. Path A, with the shared-flag coarse term and its four-site coarse preparation, delivers an excellent unfiltered input

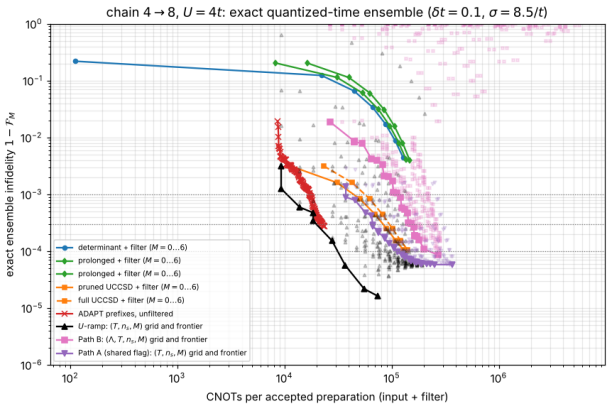


Figure 5: Cost versus accuracy for the chain $4 \to 8$ at $U = 4t$ with the exact quantized-time ensemble: lines with markers are the fixed inputs as M grows from 0 to 6 and the unfiltered ADAPT prefixes; faint symbols are every point of the (T, n_s, M) grid of the ramp, the (Λ, T, n_s, M) grid of Path B and the (T, n_s, M) grid of Path A (both coarse states), with their lower envelopes drawn; dotted lines mark the targets of Table 2.

(2.7×10^{-4} at $T = 12/t$ with 128 steps, 8.9×10^{-4} with 64) and costs 2.4–3.1 times the ramp at the three targets, instead of the factor sixteen that a ladder per rotation would have charged; it is the cheapest of the routes that start from the coarse eigenstate by a wide margin. The pruned UCCSD state costs 3.3–3.6 times the ramp and does not reach 10^{-4} within six cycles; the full template a further 25%. Path B costs 6–9 times the ramp: at any Λ its ceiling of $(t/\Lambda)^2$ forces cycles, and the Trotter step it needs grows with Λ. The prolonged state and the bare determinant reach 4×10^{-3} after six cycles and none of the targets. The determinant and the prolonged coarse eigenstate start from the same place — the same overlap (0.78 against 0.80) and the same Δ_{eff}, since both lie in the symmetry sector of the target — so the ramp pays only fine-chain steps for the same transport that Path A pays fine steps, the coarse term and the coarse preparation for; the remaining question, taken up next, is what happens when the coarse eigenstate is the better start. The costs are per accepted preparation at the operating point of each route; the setup costs — the classical optimization and its measurements for the variational states, the calibration of $(T, \mu_{\mathrm{buf}}, n_s)$ or (T, Λ, n_s) for the adiabatic inputs, the energy scan that locates E_{obj} — are not included and are amortized over the preparations that use them. The grids are disclosed and comparably dense for the three adiabatic routes,

Table 2: Minimum CNOTs per accepted preparation reaching a target ensemble infidelity for the chain $4 \to 8$ at $U = 4t$ (sixteen system qubits), per route, from the exact quantized-time ensemble of the Trotterized filter (second order, $\delta t = 0.1/t$, $\sigma = 8.5/t$, $E_{\rm obj}$ the Trotter-effective ground energy; a cycle costs 2.0×10^4 CNOTs on average). The adiabatic routes are minimized over (T, n_s, M) — for Path B also over Λ, for Path A and B also over the two charged coarse states — the fixed inputs over M, and ADAPT over its prefixes; the operating point is given in parentheses as (T, n_s, M), (Λ, T, n_s, M), (operators, M) or (M); a dash: not reached within six cycles. Path A is costed with the shared-flag coarse term (156 CNOTs per application) and includes the four-site coarse preparation; the variational inputs exclude their training.

route	10^{-3}	3×10^{-4}	10^{-4}
U-ramp from the determinant	1.4×10^4 (12, 96, 0)	2.8×10^4 (16, 192, 0)	3.7×10^4 (24, 256, 0)
ADAPT-VQE	1.7×10^4 (149, 0)	2.3×10^4 (194, 0)	7.9×10^4 (177, 3)
Path A (shared flag)	3.7×10^4 (12, 64, 0)	6.7×10^4 (12, 128, 0)	1.1×10^5 (16, 128, 2)
UCCSD pruned (98) + filter	5.1×10^4 (2)	9.0×10^4 (4)	—
UCCSD full (198) + filter	6.3×10^4 (2)	1.0×10^5 (4)	—
Path B	1.2×10^5 (16, 24, 512, 2)	1.6×10^5 (16, 24, 512, 4)	2.7×10^5 (16, 32, 1024, 6)
prolonged coarse state + filter	—	—	—
determinant + filter	—	—	—

but a grid is not a global optimum; every minimum of Table 2 is attained away from the upper edges of its grid (longer times, more steps or more cycles would not lower it), which is the sense in which the rankings are stable.

6 Eight to sixteen sites at the circuit level

The comparison of Sec. 5.3 is at one size, where the free-fermion determinant has the same overlap and the same $\Delta_{\rm eff}$ as the prolonged coarse eigenstate; the case for refinement rests on the regime where the coarse eigenstate is the better start. Here the chain $8 \to 16$ at $(2, 2)$, $U = 4t$ ($784 \to 14\,400$ states, thirty-two qubits) is taken to the circuit level with the same accounting. Every product-formula factor of the fine chain is applied through the Kronecker structure of Eq. (2) as a pair of 120×120 exponentials, the embedded coarse term of Path A exactly through the 784-dimensional coarse space, and the variational comparators with sparse generators; the Trotter-effective ground state of the fixed-step filter, which cannot be obtained by diagonalizing the 14 400-dimensional step unitary, is obtained by iterating the quantized-time filter itself from the exact ground state (details and validation against the dense eight-site machinery in the Supplemental Material [20]).

The inputs. The prolonged coarse ground state now starts ahead of the determinant, $p_0 =$ 0.869 against 0.716, with the same $\Delta_{\rm eff} = 0.138t$ for both (the determinant lies in the same symmetry sector), while the sector gap is $0.051t$ and the sector gap along the ramp falls from $0.165t$ to $0.051t$. Figure 6(left) shows the exact-slice transports. The ramp needs $T = 32/t$ for $1-p = 4.1 \times 10^{-3}$ and $T = 64/t$ for 8.1×10^{-4} (1.1×10^{-3} at $48/t$), against $T = 8/t$ for $p = 0.9924$ ($T^* = 16/t$, $p = 0.99931$; 0.99995 at $32/t$) along Path A with $\mu_{\rm buf} = 2t$; a larger shift does not help at this size ($p(16) = 0.99879$ and 0.99792 at $\mu_{\rm buf} = 4t$ and $8t$), because it raises the norm of the generator without widening a gap that is already set by $\Delta_{\rm eff}$. Path B sits at its ceiling from $T = 16/t$ on, $p = 0.99492$ and 0.99531 at $T = 16/t$ and $32/t$ for $\Lambda = 8t$ (0.99500 and 0.99605 for $\Lambda = 16t$), the same few 10^{-3} as at eight sites. The product formulas need $n_s = 128$ second-order steps for the ramp at $T = 16/t$ ($p = 0.96927$, 39 392 CNOTs) and 256 at $T = 32/t$ (0.99556, 78 304 CNOTs; with 32 steps, $\delta t = 1/t$, the ramp overlap collapses to 0.005), 256 for Path B at $\Lambda = 8t$, $T = 16/t$ (0.99476, 77 856 CNOTs plus the coarse preparation), and for Path A with the coarse term split into its conditioned groups 64 steps at $T = 8/t$ (0.99228 against the exact 0.99237) and 128 at $T = 16/t$ (0.99930 against 0.99931), i.e. 1.2×10^5 CNOTs at $T = 16/t$ with the shared-flag coarse term, of which 8.5×10^4 in the coarse term. The single-layer UCCSD ansatz on sixteen sites has 1 022 parameters, costs 2.4×10^5 CNOTs and reaches only $1 - F = 7.0 \times 10^{-3}$ (against 3.1×10^{-3} with 198 parameters at eight sites,

a deterioration by a factor 2.2); pruning to the 255 amplitudes above 10^{-3} keeps that accuracy at 6.2×10^4 CNOTs. Figure 6(right) shows ADAPT-VQE with the same pool: at eight sites the adaptive ansatz passes the single-layer accuracy at 66 operators and reaches 10^{-3} at 149 and 3×10^{-4} at 194 operators (a third of the template's operators, chosen adaptively and with repeats, do better than the full template); at sixteen sites it needs 187 operators (4.8×10^4 CNOTs) to reach 10^{-2} and has not reached 3×10^{-3} after 200. The convergence is slow because the ground state spreads over many small amplitudes rather than a few large ones, the opposite of the pairing case of paper I, where five operators sufficed at eight qubits.

The filter at thirty-two qubits. With $\sigma = 5/\Delta_{\text{eff}} = 36/t$ and $\delta t = 0.1/t$ a cycle takes 289 steps on average, 1.8×10^5 CNOTs, nine times the cycle of Sec. 5.3: 4.3 times more steps, since Δ_{eff} is 4.3 times smaller, and 2.1 times more per step. The Trotter-effective ground state, obtained by iterating the quantized-time filter itself, has an eigenvector residual $\|U_{\delta t}\psi - e^{-iE_{\text{eff}}\delta t}\psi\| = 2\times10^{-15}$; its energy lies $4.9\times10^{-3}t$ above E_0 and the floor of the filter is 3.7×10^{-5}. The cost of the filter therefore scales as $1/\Delta_{\text{eff}}$ per cycle at fixed δt, and an input that is cheap but needs several cycles pays this price in full at every size.

Matched comparison. The search of Sec. 5.3 is repeated with forty shared time sets (the 14 400-dimensional step cannot be diagonalized, so the ensemble is sampled; bootstrap relative standard errors are below 15% for every operating point used, and the selection of the cost-minimizing operating point of each route is itself bootstrapped over the schedules): the ramp over $T = 32$–$128/t$ and $n_s = 256$–$1\,024$, Path B over $\Lambda = 8t$, $16t$, $T = 16$, $32/t$ and $n_s = 256$, 512, Path A over $T = 8$–$32/t$ and $n_s = 64$–512, with the coarse eigenstate of the hierarchy prepared by the cheapest eight-site unfiltered ramp of Table 2 below 3×10^{-4} ($T = 16/t$, 192 steps, 1.8×10^{-4}, 27 760 CNOTs) or below 10^{-4} ($T = 24/t$, 256 steps, 5.6×10^{-5}, 36 976 CNOTs), prolonged ($p_0 = 0.8689$ and 0.8688 against 0.8692 for the exact coarse eigenstate) and transported, and charged to every input that starts from it; the determinant, the prolonged states, the pruned UCCSD state and the ADAPT prefixes complete the set. Table 3 gives the minima and Fig. 7 the envelopes.

The picture at thirty-two qubits differs from eight sites in two respects and agrees in a third. First, the variational routes have fallen behind: the adaptive ansatz is still the cheapest input at 10^{-2} (4.8×10^4 CNOTs, 0.6 times the ramp), but it needs two, four and six filter cycles at the tighter targets and then costs 5, 5 and 7 times the ramp; the pruned UCCSD state behaves the same way (5, 4 and 6 times). Second, the unfiltered adiabatic routes dominate from 3×10^{-3} down, and a filter cycle, at 1.8×10^5 CNOTs, is worth more than any of them: the ramp reaches 3×10^{-3} at $T = 48/t$ with 256 steps (7.8×10^4 CNOTs), 10^{-3} and 3×10^{-4} at $T = 48/t$ and $64/t$ with 512 steps (1.6×10^5), and Path A reaches 10^{-3} at $T = 16/t$ with 128 steps (1.5×10^5, of which 2.8×10^4 is the eight-site coarse preparation and 8.5×10^4 the coarse term) and 3×10^{-4} at $T = 32/t$ with 256 steps (2.8×10^5), without a cycle in either case. Third, as at eight sites, the projector path and the ramp are the two cheapest routes at the tight targets and the ramp is ahead or tied: at 10^{-3} the two coincide (1.5 against 1.6×10^5 CNOTs, a tie within the grids), at 3×10^{-3} and 3×10^{-4} the ramp is cheaper by a factor 1.9 and 1.8, and at 10^{-2} by 15%. The coarse eigenstate does start ahead of the determinant here ($p_0 = 0.87$ against 0.72) and Path A does transport faster ($T = 16/t$ against $48/t$ for 10^{-3}), but the ramp's step costs 304 CNOTs where Path A's costs $304 + 2\times332$, and the three-fold longer ramp with the cheaper step ends up at the same price. Path B, costed from the same charged coarse state, is 1.4–5.9 times the ramp: its ceiling forces cycles at every target below 10^{-2}. The prolonged state and the determinant, filtered without any transport, reach only 2.3×10^{-3} and 7.4×10^{-3} after six cycles at over 10^6 CNOTs. The ramp at $T = 16/t$, the cheapest input at eight sites, is no longer on the frontier at any target.

7 Discussion

What the benchmark shows. At sixteen and at thirty-two qubits the cheapest route to every target from 3×10^{-3} down is an unfiltered adiabatic ramp from the free-fermion determinant, and the projector path of resolution re-

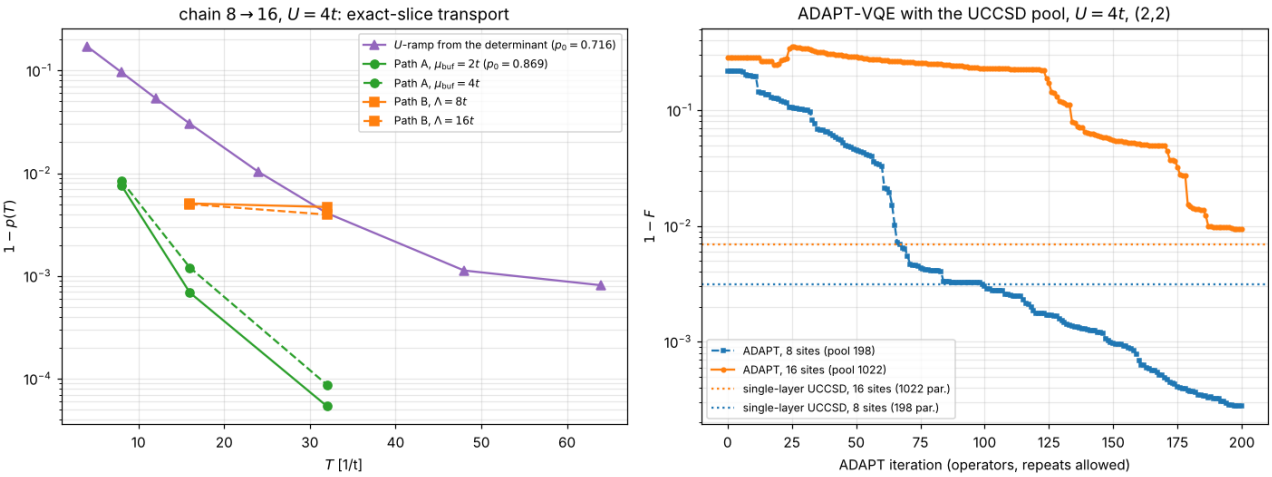


Figure 6: The chain $8 \to 16$ at $(2,2)$, $U = 4t$. Left: $1 - p(T)$ with exact slices for the ramp from the free-fermion determinant, Path A with $\mu_{\mathrm{buf}} = 2t$ and $4t$ and Path B with $\Lambda = 8t$ and $16t$. Right: ADAPT-VQE with the UCCSD pool at eight and at sixteen sites, infidelity against the number of operators, with the single-layer UCCSD values as horizontal lines.

Table 3: Minimum CNOTs per accepted preparation reaching a target ensemble infidelity for the chain $8 \to 16$ at $U = 4t$ (thirty-two system qubits), per route: the same Trotterized filter (second order, $\delta t = 0.1/t$, $\sigma = 36/t$, E_{obj} the Trotter-effective ground energy, forty shared time sets; a cycle costs 1.8×10^5 CNOTs on average), the adiabatic routes minimized over (T, n_s, M) — for Path B also over Λ, for Path A and B also over the two charged coarse states — the fixed inputs over M and ADAPT over its prefixes; the operating point in parentheses as in Table 2. Path A is costed with the shared-flag coarse term (332 CNOTs per application); every input that starts from the coarse eigenstate includes its eight-site preparation. In 400 bootstrap resamples of the schedules the listed operating point remains the cheapest of its route in every resample for the unfiltered adiabatic points and in 61–100% of them for the filtered inputs (72–100% for the variational inputs); the 5–95% ranges of the resampled minimum costs are given in the Supplemental Material [20].

route	10^{-2}	3×10^{-3}	10^{-3}	3×10^{-4}
U-ramp from the determinant	7.8×10^4 (32, 256, 0)	7.8×10^4 (48, 256, 0)	1.6×10^5 (48, 512, 0)	1.6×10^5 (64, 512, 0)
Path A (shared flag)	9.0×10^4 (8, 64, 0)	1.5×10^5 (16, 128, 0)	1.5×10^5 (16, 128, 0)	2.8×10^5 (32, 256, 0)
ADAPT-VQE	4.8×10^4 (187, 0)	3.9×10^5 (187, 2)	7.9×10^5 (187, 4)	1.1×10^6 (187, 6)
UCCSD pruned (255) + filter	6.2×10^4 (0)	4.1×10^5 (2)	5.9×10^5 (3)	9.8×10^5 (5)
Path B	1.1×10^5 (8, 16, 256, 0)	2.7×10^5 (8, 16, 256, 1)	5.3×10^5 (16, 32, 512, 2)	9.2×10^5 (16, 32, 512, 4)
prolonged coarse state + filter	8.1×10^5 (4)	1.1×10^6 (6)	—	—
determinant + filter	1.2×10^6 (6)	—	—	—

finement, compiled with its coarse term conditioned on one shared flag qubit, is the cheapest of the routes that start from the coarse eigenstate: 2.4–3.1 times the ramp at sixteen qubits, tied with it at 10^{-3} and within a factor 1.9 elsewhere at thirty-two. The reason the gap closes is the one the refinement proposal predicted — the coarse eigenstate starts ahead of the determinant at thirty-two qubits and the projector path transports in a third of the time — but the reason it does not reverse is the cost of the step: the ramp pays 304 CNOTs per step where Path A pays 304 plus two applications of the coarse term at 332, so a three times longer ramp with the cheaper step ends at the same price. The variational routes are the cheapest inputs only where they need no filter cycle (10^{-3} and 3×10^{-4} at sixteen qubits for the adaptive ansatz, 10^{-2} at thirty-two for ADAPT and the pruned template); below that they pay for cycles, and a cycle, whose cost grows as $1/\Delta_{\mathrm{eff}}$, is worth more as extra evolution of an adiabatic input than as a correction to a variational one at every target tested. The comparison was run on equal terms — the same filter, the same cost model, a joint search over time, steps and cycles for every route with the actually prepared coarse state charged and propagated — and three of its conclusions are specific to what was tested: the costs are amortized logical CNOTs per accepted preparation on an ideal

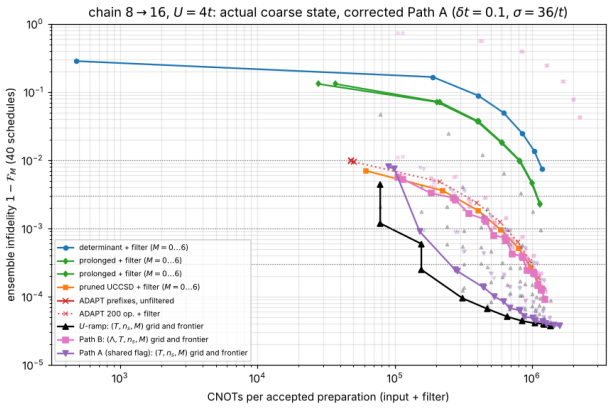


Figure 7: Cost versus accuracy for the chain $8 \to 16$ at $U = 4t$ (forty shared time sets): the fixed inputs as M grows from 0 to 6, the unfiltered ADAPT prefixes and the 200-operator state through the filter, and every point of the ramp, Path-B and Path-A grids with their lower envelopes; dotted lines mark the targets of Table 3.

device, without the rejected attempts, the setup costs or any noise; the variational comparators are the UCCSD template, its pruning and the ADAPT-VQE grown from the UCCSD pool, and the Hamiltonian-variational ansatz, whose layers are the Trotter steps of the ramp with free coefficients, is the natural competitor we have not run; and the family is the dilute $(2, 2)$ sector, where the free-fermion determinant is a good start and the U-ramp an easy path. Two lessons are inherited from paper I and sharpened. A coarse term conditioned on every antibonding qubit is never affordable, but the same term conditioned on one flag qubit, at the price of $c-1$ clean work qubits, costs little more than a fine step; that compilation, or the two-local path with its ceiling, is the form of the projector path worth running. And the figure of merit is the minimum cost at fixed accuracy per accepted preparation with the cycles included and every parameter searched: an input that "converges to the floor" in a fixed number of cycles is not thereby cheaper than one that reaches the target without them.

Which gap matters. The quantity that controls the refinement is the symmetry-protected effective gap Δ_{eff}, not the sector gap; this is why T^* does not grow with U from 0 to $8t$ at $4 \to 8$ and why the first tested $16/t$ still suffices at $8 \to 16$ with a sector gap of $0.05t$, why the sufficient time of the adiabatic theorem is a closer (though still remote) estimate when evaluated with the symmetry-resolved gap, and why Δ_{eff} is the right width for the filter. The accessible gap itself shrinks with the size ($0.59t \to 0.14t$; at $U = 0$ and fixed particle number the lowest accessible even-singlet excitation is $2t[\cos\frac{\pi}{L+1} - \cos\frac{3\pi}{L+1}] \sim 8\pi^2 t/(L+1)^2$), so symmetry excludes the forbidden states but does not supply an asymptotically finite gap, and the timing is a favorable observation over the tested cases, not a scaling law. Paper I found that its scan of the refinement time against the shift-controlled gap did not identify a power law; the lesson here is that any such scan must be made against the gap in the symmetry sector of the tracked state, with the response norm held fixed, before an exponent can be read off. The zero-overlap plaquette is the converse and a warning for any refinement, fusion or nucleation scheme: the coarse state must carry the quantum numbers of the target, which the ED diagnostics of Sec. 2 — or, on a device, measurements of $\mathbf{S}^2$ and of the point-group characters — can check before any gate is spent.

Lattice versus basis refinement, and coarse accuracy. The lattice refinement is structurally cleaner than the basis refinement of paper I: a Galerkin coarse model in closed form, a self-similar hierarchy that starts from a single basis state, a two-local Hamiltonian with $O(L_f)$ Pauli strings of weight two, a prolongation of $O(N_c)$ gates. Against this stand the conditioned coarse term, Path B as its two-local replacement with a $(t/\Lambda)^2$ ceiling and a step count that grows with Λ, and three to ten times more Trotter steps than the pairing path for the same adiabatic time, because the shifted generators have larger norms. Composing steps with refined inputs and starting from a variational coarse state show the same ceiling: an admixture of other coarse eigenstates refines to other fine eigenstates and caps the final overlap at roughly the input fidelity, so the coarse stage must be as accurate as the target, or the filter must pay for the difference; the filled dimer, which is exact, is from this point of view the best coarse state the hierarchy has.

Outlook. At fixed U/t the hierarchy is self-similar, so refinement does not by itself reach strong coupling; a path that raises U with the resolution, or a refinement at fixed filling, are the options. In one dimension the dimer-seeded

hierarchy and the nucleation of a chain from a dimer [17] start from the same state and reach the same target by refining resolution versus growing the domain, and the Hubbard chain is the natural place to compare the two at matched gate count; the optimized variants of either (a warm-started variational path) are the obvious way to cut the 10^2–10^3 Trotter steps found here. The refinement and filter parameters are calibrated against the exact answer; a device-level protocol (an energy scan for E_{obj}, acceptance statistics for σ and δt, a convergence test in T and n_s on the measured energy) and a comparison at matched shot budget including the training measurements of the variational routes are the missing pieces for a hardware run, for which the $4 \to 8$ chain and the $(1, 1)$ plaquette (thirty-two qubits, 256-dimensional sector) are within reach of current devices.

Acknowledgments

The work of C.G. and G.L. acknowledges funding from the European Union's Horizon Europe research and innovation program under the Marie Skłodowska-Curie Grant Agreement No. 101126636. M.H.-J. is partially supported by U.S. National Science Foundation (NSF) Grants PHY-1404159 and PHY-2310020. The authors acknowledge Anthropic's Claude for assistance with literature search and coding.

Code and data availability

All results are produced by the notebook `notebooks/ResolutionRefHubbard.ipynb`, built by `scripts/build_notebook_hubbard.py` on the module `pairinglib.hubbard` (lattices, sector ED, block prolongation, Galerkin coarse model, both paths, exact and Trotterized refinement, the ensemble rodeo formula and its Trotterized version, the symmetry diagnostics, the sector-level UCCSD and ADAPT-VQE, the Kronecker-structured product formulas of the sixteen-site study and the gate counting), which is covered by the test suite of the repository [18]; the figures are extracted from the executed notebook, and the Supplemental Material [20] and the notebook together document every number quoted. The ED reference code (Hubbard and t–J models; chain, square and checkerboard lattices) is the notebook `Hubbard.ipynb` [22], whose conventions the module follows.

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